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الحياة الطيبة

Software Requirements & UML Specification

Client review draft | Version 0.2 | 29 September 2026

A complete single-door MVP for children aged 10-14 and their mothers, with independent learning on separate devices and an extensible content-management workflow.

Review status	Purpose
Draft for review	Confirm scope and behavioral requirements before UI design and detailed implementation.
Confirmed and proposed items	Confirmed product direction is distinguished from proposed mechanisms and unresolved decisions.

Includes the SRS, twelve detailed use cases, logical architecture, ten UML diagrams, and requirements-to-acceptance traceability.

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How to read this document

Start with the executive brief and decision register. Review the SRS and use cases before treating the models as an implementation baseline. The UML section uses larger pages where necessary to keep technical labels readable.

Status notation

D - derived: operationalizes a confirmed product decision. **P - proposed:** mechanism or scope awaits agreement. **Must / Should:** priority within the proposed MVP. Normative wording makes a draft requirement testable; it does not mean it has already been approved.

01 Executive brief

Product direction

Hayah Tayeba helps children aged 10-14 understand a value, explore responses through a story, and try a small action in everyday life. Mothers have both their own learning journey and optional ways to support their children. Each uses a separate device. The mobile client will use Expo React Native.

The first and only complete MVP door is «باب الذكر», replacing the former good-character/anger journey. The full route introduces meaning, practical entry, continuing practice, and reflection. Children create their character before entering «خريطة الحياة الطيبة». Arabic is the release language, with a locale-neutral structure for reviewed English editions later. Completion does not certify worship, emotional change, or spiritual growth.

MVP boundary

- One complete Dhikr door with multiple learning stages, corresponding mother content, and optional shared activities.
- Independent child progression after the proposed account setup.
- Saved learning progress and recovery from ordinary connection interruptions.
- An admin workflow for adding content using supported templates.
- Explicit content review, publication, revisions, and withdrawal.

Decisions required before implementation

- Adult sign-in, initial child setup, and device pairing.
- Exactly what the mother may see about the child's activity.
- Content authorship, review ownership, and the mother's first lesson.
- Launch platforms, distribution countries, and connectivity expectations.
- Data-retention policy, operating ownership, and release quality targets.

Delivery status

This document specifies a proposed solution. Application development, final curriculum production, religious-source verification, and product acceptance testing have not been completed. The supplied presentation, expanded project notes, and revised Dhikr concept are conceptual inputs rather than an approved finished curriculum.

Review approach

Agree the behavioral scope first, then select backend technologies and physical data/API designs. UI prototyping can still reveal issues that require revisions to the specification. No price, delivery date, hosting provider, or launch commitment is established by this draft.

02 Client revision and Dhikr scope

Change summary

The client changes the audience to **10-14**, expands the first release to **one complete door**, confirms **Arabic first with an English-ready content structure**, and places **character creation before the locked-door map**. The product owner has explicitly confirmed **«باب الذكر»** as the first door, replacing the former anger journey inside the good-character door.

Mother's own journey, optional mother participation, independent child learning, separate devices, Expo React Native, and periodic admin content updates remain in scope. The new child concept does not supply a finished mother curriculum.

Source interpretation

The new Dhikr note is a content/design proposal. Its English dialogue is reference copy to adapt into Arabic, not a bilingual-launch instruction. Its example XP amounts, seven-day window, ten-second practice, and ten-minute opening are not finalized engineering targets. Written and voice reflection modes require an explicit scope and data-handling decision. Religious wording and learning-effectiveness claims still require editorial review.

The written/voice-reflection choice has been raised with the product owner and remains pending. v0.2 documents that request instead of silently retaining the previous fixed-choice-only proposal or automatically introducing audio uploads.

Full door structure proposed for the MVP

Stage	Child experience	Completion meaning
1. Character and entry	Select avatar, nickname, appearance, color, and badge; enter «خريطة الحياة الطيبة»; open the Dhikr introduction.	Setup and entry participation, not an assessment.
2. Discover meaning	A familiar distracted-study situation leads into choices, explanation, and an editorially reviewed religious source.	Explore the meaning through authored scenes and feedback.
3. Try a practice	A short guided pause and approved Dhikr example; feedback may include little or no change in feeling.	Participation, never emotional success or worship quantity.
4. Choose a moment	Discover everyday contexts and create one when/then plan.	A chosen practical intention; no proof of real-world worship.
5. Continue	Suggested seven-day journal with optional tried/remembered/remembered-independently check-ins.	Practice history distinct from digital learning completion; pauses do not erase progress.
6. Pause and reflect	Consider when it was easiest, hardest, and where to continue.	Optional response, explicit end of the learning route, and invitation to continue.

Each stage must contain real reviewed content in the MVP. “Complete door” does not mean unlocking every future door or proving lifelong habit formation. Final stage count, mission count, duration, and mother's corresponding lessons require editorial sign-off.

Disposition of the supplied concepts

Concept	Treatment in v0.2	Engineering consequence
Child as hero / custom avatar	Confirmed setup direction	Persist preset appearance/color/badge choices; keep separate from adult account provisioning.
Map and locked doors	Confirmed direction	Separate progression locks from unavailable future previews; only Dhikr has a complete curriculum.
Door, Discover, Missions, Garden, Quiet Place	Proposed regions inside Dhikr	Governed stage/region metadata and reachable prerequisites; art/names remain editable.
Notification story and choices	Included as a content pattern	Authored branching feedback; no punitive right/wrong framing.
Religious text introduced through story	Included with editorial review	Record sources and translation review; no unverified quotation auto-publication.
Short practice and emotional check-in	Included as a proposed interaction	No required improvement in mood; feedback does not claim treatment or guaranteed calm.
Find three moments, choose one	Proposed practice activity	Approved context catalog and when/then plan; three is an editable instruction, not a universal system constraint.
Seven-day journal and pauses	Proposed continuation design	Calendar-aware, duplicate-safe check-ins; mandatory seven-day gating remains OQ-13.
Learning XP	Proposed, optional MVP capability	Reward ledger, one eligible award per event, no worship-count or emotion scoring.
Growing tree / discovery-to-practice progression	Proposed visual representation	Derived from participation states; no claim to measure faith or prove a habit.
Omar, Mariam, Youssef, Lina	Illustrative characters	Names, personalities, scenes, and translations are authored content, not hard-coded accounts.
Secret heart/gratitude/difficult-moment doors	Proposed Dhikr subregions	They do not add complete independent curriculum doors to the MVP. Only reachable learning prerequisites may unlock them.
End-of-journey reflection in text or voice	Requested concept; modality decision pending	Storage, privacy, microphone handling, retention, and deletion depend on selected modes. No transcription/AI interpretation implied.
First ten minutes	Suggested opening storyboard	Not a fixed session-duration requirement; final pacing is tested with ages 10-14.
Remembering moments map	Proposed practice feature	Save personally chosen contexts without GPS, passive sensing, or worship surveillance.

Concept	Treatment in v0.2	Engineering consequence
English wording in the note	Reference only	Arabic release copy; later reviewed English editions share canonical content/progress IDs.

Arabic copy direction - illustrative, not final curriculum

- Map: «خريطة الحياة الطيبة».
- Main door: «باب الذكر».
- Example practice intention: «لما أستعد للنوم، هفتكر الذكر اللي اخترته».
- After a pause: «تقدر تكمل من هنا. الرحلة فيها وقفات».
- Future preview: «هذا الباب سيضاف لاحقا».
- Reflection can be skipped: «مش حابب أجاوب دلوقتي».

Dialect, exact devotional wording, religious explanation, and age suitability require review by the project's appointed content owner. Ordinary anger and distraction may appear as situations inside Dhikr; that does not restore the former good-character door as a second MVP door.

Main unresolved choices

1. Written/voice/fixed-choice reflection modes and storage/access policy (OQ-14).
2. Whether seven days is only an invitation or a compulsory completion condition (OQ-13).
3. Exact reward values and unlock rules (OQ-15).
4. The complete mother curriculum and editorial/asset-production responsibilities (OQ-03/OQ-07).

Earlier open decisions about authentication, launch platforms, offline access, distribution, and retention also remain active. The earlier PDF is preserved as v0.1; this revision replaces it for current review.

03 Decisions and open questions

Confirmed product direction

ID	Confirmed decision
C-01	Children are aged 10-14.
C-02	The mother has a personal journey and a role supporting her child.
C-03	Children can start and complete their experience independently; mother participation is optional.
C-04	The only complete MVP door is «باب الذكر». The previous good-character/anger journey is superseded as the first release.
C-05	Light fantasy setting with everyday situations and dialogue.
C-06	Mobile client uses Expo React Native.
C-07	Mother and child use separate devices.
C-08	Admins can periodically add new content.
C-09	Deliver one complete door: meaning, entry, practical application, continuation, reflection, and a clear end to its learning route.
C-10	Existing inputs are the presentation, original notes, and the new Dhikr concept. Final production content and application code are not supplied.
C-11	Hold UI design and prepare SRS/UML first.
C-12	Arabic is the release language; the content structure must permit English later without redesign.
C-13	The child creates a character before entering the map, choosing avatar, name, appearance, color, and a personal badge.
C-14	Present «خريطة الحياة الطيبة» with doors and visible locks; examples include Dhikr and Allah's companionship.

Proposed defaults - awaiting confirmation

ID	Proposal	Reason and limit
P-01	Mother authenticates; child uses a linked profile and a device session without a child email or phone.	Provides separate-device access without requiring child contact details. Identity provider is undecided.
P-02	Temporary, single-use pairing code with explicit mother confirmation of the pending device.	Makes linking understandable while reducing accidental linking. Detailed timing/rate limits are provisional.
P-03	One responsible mother account per child in MVP; one mother may manage multiple children.	Bounds ownership and authorization. Additional guardians and ownership transfers are deferred.

ID	Proposal	Reason and limit
P-04	Mother's overview exposes topics and completion, not individual choices or personal reflections.	Separates support from detailed behavioral surveillance. This visibility boundary requires confirmation.
P-05	Reflection input mode is being revised: the client concept requests writing or voice; the user has not yet selected MVP modalities.	The old fixed-choice-only proposal is no longer treated as settled. Text/voice retention, backup, and access need a decision before implementation.
P-06	Templates support stories, video/animation, choices with feedback, short challenges, real-life practice plans, check-ins, and reflection prompts.	Dhikr is the first authored door; no general-purpose game engine is implied.
P-07	Admin console is a separate desktop web application.	Easier authoring and preview. Framework is undecided.
P-08	One admin role initially; review and publication are explicit actions, possibly by the same authorized person.	No unrequested multi-role approval bureaucracy. Independent editorial review can be added if the team requires it.
P-09	Immutable published content revisions; attempts bind to a revision.	Protects in-progress sessions and historical completion. Withdrawals need explicit handling.
P-10	Online content access, with durable local progress and a retry queue; no full offline catalog/download manager.	Handles ordinary interruptions without claiming complete offline operation.
P-11	No punitive streaks, leaderboards, spirituality scores, or mother sign-off to advance. Optional learning XP and visual growth acknowledge participation only.	Exact XP values and reward rules in the source are examples, not finalized acceptance thresholds.
P-12	Locale-neutral IDs and versioned Arabic/English language editions; Arabic is complete for launch. Child copy can use accessible Arabic appropriate to ages 10-14.	Arabic-first and future bilingual structure are confirmed; dialect and translation workflow are proposed.
P-13	One backend deployment with modular responsibilities, transactional data, and separate media storage.	A starting architecture, not a selected provider or mandatory infrastructure stack.
P-14	Use supported illustrated scenes and media for the complete Dhikr route. Rich custom mini-games and full animation production are separately scoped.	The supplied concepts do not imply that all media assets or game mechanics already exist.
P-15	One active child learning-device session per profile; replacement revokes the prior session.	Simplifies cursor synchronization; multiple children can still use separate devices. Simultaneous devices for the same child require a later conflict policy.

Open decisions

The timing column identifies when each decision is needed for the next delivery stage.

ID	Question	Needed before	Working treatment
OQ-01	Accept linked child profiles and mother-approved device pairing? What adult sign-in method?	Auth implementation	P-01/P-02 modeled provisionally.
OQ-02	What exactly may a mother see about her child's activity?	Data schema and API baseline	P-04/P-05 minimize disclosure.
OQ-03	Who authors and reviews educational/religious material and media usage rights?	Production content publication	Review fields exist; no reviewer authority is invented.
OQ-04	Is Android first, iOS first, or both at launch? Minimum OS/device/browser support?	Build and QA plan	Expo chosen; platform release scope remains open.
OQ-05	Must complete journeys work offline, or is interruption recovery enough?	Storage/media design	P-10 is the draft scope.
OQ-06	Is the first release private pilot or public distribution, and in which countries?	Launch/privacy policy	No legal compliance determination made.
OQ-07	What is the mother's Dhikr lesson and personal practice?	Curriculum acceptance	Her independent path remains in scope; the new child concept does not supply a finished mother curriculum.
OQ-08	Separate content adaptations for 10-11 and 12-14, or one initial journey tested across both?	Content production	One initial version proposed; usability reviewed across ages.
OQ-09	Is initial adult setup acceptable before a child can start independently?	Onboarding baseline	Independence during learning is confirmed; zero-setup guest mode is not.
OQ-10	Monetization, budget, expected pilot size, media volume, and availability target?	Hosting and operating plan	No ads, payments, or subscription system included in draft MVP.
OQ-11	Account deletion, retention periods, backup retention, consent notices, and support owner?	Public release	Draft deletion behavior is specified; periods remain TBD.
OQ-12	How should editorial safety withdrawals affect unfinished sessions?	Content lifecycle implementation	Stop withdrawn content, preserve records, offer another available route.
OQ-13	Is seven days a suggested practice window or a compulsory completion period?	Progress rules	Proposed: seven-day invitation, no consecutive-day requirement or lost progress. Digital completion and ongoing practice remain separate.
OQ-14	Which reflection modes launch: fixed choices, written text, and/or voice? Where are they stored?	Reflection implementation	Pending user response. No text or audio storage mechanism is approved by the source alone.

ID	Question	Needed before	Working treatment
OQ-15	Which events award XP and unlock regions, and what are the reward values?	Reward implementation	Proposed: participation only, idempotent grants, no worship-count or emotion scoring.
OQ-16	Which future-door previews appear, and when is a lock actually unlockable?	Catalog baseline	Dhikr is the only full door; «الله معية» may be a clearly labeled future preview.

Items not yet adopted as requirements

“بلدة طيبة” as a world name; exact palette; a particular mascot; unrestricted avatar uploads or a full character editor; a fixed lesson duration; collectible artwork; push notifications; punitive streak rules; automatic content personalization; any AI tutor; any named backend/database/cloud vendor. Exploratory visual concepts do not constitute approved product requirements.

Tradeoffs to revisit

- A curated route is easier to author than arbitrary branching. The initial editor should allow bounded choices and explicit feedback, not a general game engine.
- Detailed parent visibility can support intervention, but can also discourage honest reflection. Resolve the boundary explicitly before storing answers.
- Dynamic content reduces app releases only for supported templates and schema versions. New client capabilities need a client update.
- One shared age range lowers production cost, but usability testing may show that reading level, pacing, and visual style need variants.
- Pausing UI design is a sequencing choice. A small flow prototype may still be needed to validate pairing and child comprehension before finalizing requirements.

04 Software requirements specification

1. Purpose and interpretation

Specify the behavior of the first usable product and its content-management capability before visual design and backend technology selection. This is a tailored SRS, not a claim of conformance to a particular standard.

D (derived) means a requirement directly operationalizes a confirmed product decision. **P (proposed)** means its mechanism or scope needs product approval. Both remain part of a draft. “Shall” makes the proposed baseline testable; it does not imply approval. Sources C-xx and P-xx are defined in Section 03.

Priority: **Must** = required for the proposed MVP; **Should** = desirable but negotiable. Items explicitly deferred are not MVP commitments.

2. Product purpose, scope, and success

Help children aged 10-14 explore a value, practice a response, and try a small action in everyday life. Mothers pursue their own learning and optionally support their children. The first and only complete MVP door is «باب الذكر». It covers meaning, entry, practical application, continued practice, and a closing reflection. The former good-character/anger journey is superseded as the initial release.

The product shall represent learning participation and practice, not certify spiritual growth, moral worth, or emotional health. Completion is not evidence that a child has changed behavior outside the app.

Proposed MVP includes

- Separate mother and child experiences using Expo React Native.
- A proposed family-linking and device-access flow.
- One complete Dhikr door with several learning stages; corresponding mother content whose final curriculum is pending; optional shared activities.
- A lightly fantastical map with locked doors and everyday situations, preceded by child character creation; Arabic at launch and English-ready content structure.
- Supported content templates and an admin authoring/publication workflow.
- Persisted progress, resumption, duplicate-safe synchronization, and basic recovery states.
- Authorization, accessibility, editorial provenance, and operational controls appropriate to this scope.

Deferred

Custom game builder, free-roaming 3D world, generative AI/chat, social messaging, public child profiles, public user uploads, competitive rankings, payments/ads, full offline media library, teacher/school roles, multiple guardians, push campaigns, and automated adaptive learning.

Acceptance outcomes

- A linked child can finish the available journey without adult approval during learning.
- A mother can complete her own content independently of the child's progress.
- An admin can publish a second journey using supported templates without changing the client binary.
- Existing completion and in-progress sessions remain interpretable after a routine content revision.
- A network interruption does not silently erase acknowledged progress or cause duplicate completion.

Pilot engagement and learning effectiveness need observation and a separate evaluation plan. This SRS sets no invented adoption or efficacy targets.

3. Actors and boundaries

Actor	Allowed capability	Boundary
Child	Access their published journey, make choices, complete activities, resume progress.	No admin controls, adult account settings, other children's records, or public profile.
Mother	Manage her account and linked children, complete her own journey, view the permitted child summary, optionally participate.	No access to unrelated families; detailed child reflections excluded under P-04/P-05.
Admin	Author, preview, review, publish, revise, and withdraw content.	Content role alone grants no access to private learner records.

Authentication and media services are external technical dependencies, not additional human roles. A support/operator role is not specified for MVP; production ownership must be resolved before launch.

4. Functional requirements

Accounts, linking, and access

FR-01 | P / Must / P-01

The system shall authenticate mothers and admins and enforce capabilities on the server.

Acceptance: an unauthenticated request cannot retrieve family data; a mother or child session cannot publish content even if the request is crafted directly. Provider and recovery mechanism remain OQ-01.

FR-02 | P / Must / P-01, P-03, C-13

A mother shall provision a linked child profile with age in the 10-14 range; the child shall create their own character through FR-29. Child email/phone are not required.

Acceptance: out-of-range ages receive a clear message, child identity is scoped to one mother account, and exact birth date is not required.

FR-03 | P / Must / P-02

A mother shall initiate a temporary, single-use pairing challenge; the child device submits it; the mother explicitly confirms that pending device before a child session is issued.

Acceptance: expiry, cancellation, reuse, or confirmation by a different mother fails; simultaneous redemption cannot create two authorized sessions from one challenge. Pairing values are not logged.

FR-04 | P / Must / P-01, P-02, P-15

A mother shall view and revoke a linked child's device session and pair a replacement without deleting progress.

Acceptance: subsequent protected server requests from the revoked session are rejected; replacement pairing revokes the previous active learning session; replacement device receives server-confirmed progress. Unsynced progress on a lost device cannot be promised recoverable.

FR-05 | D / Must / C-02, C-07

The system shall maintain separate mother and child learning contexts across their devices.

Acceptance: the child's session cannot switch to the mother's private account; the mother's learning completion does not mark a child's step complete, or vice versa.

Child learning and progress

FR-06 | D / Must / C-04, C-05, C-09, C-14

The child shall see «خريطة الحياة الطيبة» with the full Dhikr door and explicitly labeled published previews of selected future doors.

Acceptance: the full Dhikr route is startable; teaser metadata may be visible, but draft lessons/media never leak through preview doors. A future door cannot falsely promise that more XP will open unavailable content.

FR-07 | P / Must / P-06, P-09

The client shall fetch a validated, versioned content manifest and render only supported schema/template versions.

Acceptance: unsupported content does not crash the client or advance progress; a suitable update/unavailable message appears. Existing supported content remains accessible.

FR-08 | P / Must / P-06, P-14

Story steps shall display illustrated scenes, short dialogue, and optional narration with accessible playback controls.

Acceptance: essential meaning remains available as text when audio is muted/unavailable; navigation does not depend on audio completing.

FR-09 | P / Must / P-06, P-11

Choice steps shall provide authored feedback or a consequence and permit another attempt.

Acceptance: each choice has a valid destination or feedback; no score of faith/character is inferred; trying another response does not erase existing completion. A required choice step completes after exploring at least one choice and its feedback, not after guessing an approved answer.

FR-10 | D / Must / C-03

A child shall be able to finish required learning steps without mother attendance, permission, or a shared activity.

Acceptance: decline/skip of every optional shared step still permits journey completion.

FR-11 | P / Must / P-06, P-11

Real-life activities and reflections shall support deferral or skipping.

Acceptance: “أجربها بعدين” and “مش حابب أجاوب” do not block the next step or completion. No proof of a real-world event is required.

FR-12 | P / Must / P-09, P-10

The system shall persist the last checkpoint and required-step completion for each attempt, queue unsent progress durably, and retry safely.

Acceptance: force-closing after a local save resumes at that checkpoint on the same device; duplicate event submission has one effect; the UI distinguishes pending synchronization from confirmed synchronization.

FR-13 | P / Must / P-09, P-11, OQ-13

Required digital learning steps shall produce version-bound journey completion. Completion of all required stages produces door learning completion; continued practice remains separate.

Acceptance: a missed day, unchanged emotion, optional reflection skip, or absence of the mother does not remove completed learning. A mandatory seven-day completion gate is not assumed.

Mother experience

FR-14 | D / Must / C-02

The mother shall have an independent content path with its own saved progress and a personal practice.

Acceptance: it works before, during, or after the child's journey. The first lesson must be authored and reviewed before release; it is not yet supplied.

FR-15 | P / Must / P-04, P-05

A mother shall see only the permitted summary for each linked child: topics accessed, required-step/journey completion, and authored support suggestions.

Acceptance: server responses exclude raw choice histories, optional reflection answers, spiritual scores, and unrelated-family data. Pending child-device progress appears only after successful synchronization.

FR-16 | D / Must / C-02, C-03

The system shall offer optional shared activities alongside solo routes.

Acceptance: neither device must be online simultaneously and no real-time parent/child chat or co-play session is required.

Admin content management

FR-17 | D / Must / C-08

An authorized admin shall add/edit doors and journeys and order their content.

Acceptance: a second supported journey can be created, published, and retrieved by an existing client without a code change.

FR-18 | P / Must / P-06, C-09, C-12

The editor shall support stories, video/animation media, choice/feedback, short guided challenges, practice plans, check-ins, reflection prompts, and optional shared activity templates.

Acceptance: stable locale-neutral IDs, stage/path, type, completion rule, and valid template data exist for every step; unsupported game mechanics are rejected.

FR-19 | P / Must / P-06

Admins shall associate images/audio/video with usage-rights notes, meaningful accessibility text/captions, and processing status.

Acceptance: unsafe or unsupported file types, unfinished uploads, and missing required assets cannot be published; upload alone never makes media public.

FR-20 | P / Must / P-08

Admins shall preview each audience's journey and record an editorial review before publication.

Acceptance: unresolved references, unreachable required steps, missing essential alternatives, unreviewed religious quotations, and unsupported schemas block publication with specific errors. A second distinct reviewer is not required by this draft.

FR-21 | P / Must / P-09

Publishing shall atomically expose an immutable revision and update the catalog pointer.

Acceptance: editing live content creates a draft; a failed publish exposes no partial journey; concurrent stale edits/publishes are rejected rather than overwriting newer work.

FR-22 | P / Must / P-09

Routine revisions shall preserve attempts bound to older available revisions and historical completion.

Acceptance: new attempts use the current revision; existing attempts resume their revision unless withdrawn; a future optional second journey does not undo completion of the first.

FR-23 | P / Must / P-09

An admin shall withdraw a published revision with a reason and explicit confirmation.

Acceptance: it disappears from new-start catalog results; the next online access check blocks further delivery for bound attempts and offers an explanation; historical progress remains. The app cannot guarantee immediate revocation of already-delivered offline bytes.

FR-24 | P / Must / P-08, P-09

The system shall audit authoring lifecycle changes, review, publish, and withdrawal with actor, timestamp, revision, action, and reason where needed.

Acceptance: failed actions are distinguishable from successful publication; secrets and child answers are absent from audit payloads.

Recovery and account lifecycle

FR-25 | P / Must / P-10

The client shall recover gracefully from network and media failures.

Acceptance: a visible retry and saved checkpoint remain; it does not claim that unfetched media works offline; a failed optional media load does not permanently trap a user.

FR-26 | P / Must / OQ-14

Reflection prompts shall be optional and nonjudgmental. Fixed choices are currently supported; written and voice modes remain conditional pending an explicit MVP decision and storage/access policy.

Acceptance: skipping does not block completion; no reflection payload enters progress, parent summaries, or telemetry. No microphone permission or upload is introduced merely because the source suggests voice notes.

FR-27 | P / Must / OQ-11

Mothers shall be able to request deletion of their account or a linked child profile with scope explained and confirmation required.

Acceptance: a confirmed request immediately revokes relevant sessions and schedules removal/anonymization of associated learner data; admin content authored by another account remains. Retention, completion deadline, and backup purge policy must be agreed before release.

Client revision: complete door, language, and practice

FR-28 | D / Must / C-12

The system shall ship with complete Arabic UI/content and a locale-neutral content model capable of English editions later.

Acceptance: translating a test journey changes localized text/audio/captions without duplicating door, step, choice, or progress IDs; no English authoring placeholders leak into Arabic release screens. The later English launch still requires translation, review, and LTR UI testing.

FR-29 | D / Must / C-13

After any required adult provisioning/linking, the child's first personal setup shall offer avatar, nickname, appearance, color, and badge before showing the map.

Acceptance: each choice persists after restart/re-pairing, can be edited later, and belongs to that child. Approved presets are proposed; public profiles and personal-photo uploads are not implied.

FR-30 | D / Must / C-14

Map nodes shall distinguish available, progression-locked, completed, withdrawn, and future-preview states.

Acceptance: a progression lock states its reachable learning prerequisite; a future-preview lock states that content is not yet available; withdrawal cannot be bypassed by an unlock record. The entry experience opens the only available MVP door without requiring another unreleased door.

FR-31 | D / Must / C-09

The MVP shall cover the whole Dhikr learning route: discover meaning, explore contextual stories, try a small practice, choose a daily moment, continue practice, and reach a closing reflection/end state.

Acceptance: no required stage is an empty placeholder; admins can author and order these stages, and digital door completion is distinct from an ongoing habit.

FR-32 | P / Must / Dhikr source

The child shall select a daily context and associate an approved practice using a when/then plan, and collect discovered remembering moments.

Acceptance: the plan survives resume, the same moment is not duplicated by retries, and no real-world location, microphone, or background surveillance is used to infer worship.

FR-33 | P / Must / Dhikr source, OQ-13

A practice journal shall accept optional self-reports such as tried, remembered, or remembered without a reminder.

Acceptance: a missed day is a neutral pause, not a reset; duplicate check-ins for the same plan/day update one record. Proposed seven-day windows use the chosen plan timezone; calendar/timezone rules are documented before implementation.

FR-34 | P / Should / Dhikr source, OQ-15

Optional XP and visual growth shall reward defined learning/participation events only.

Acceptance: retry/replay cannot farm the same reward; emotion changes and worship repetition counts never determine XP. Discovery/practice/habit labels describe participation, not verified spirituality; exact values remain to be agreed.

FR-35 | P / Must / C-08, C-12, C-14

Admins shall edit localized editions, stage prerequisites, preview-door metadata, and approved practice/reward configurations through validated content tools.

Acceptance: translation publication cannot alter progression IDs; missing required Arabic copy blocks release; cyclic/impossible prerequisites and unavailable-door unlock rules fail validation; user reflection content is not exposed in this editor.

5. Business rules

ID	Rule
BR-01	Optional mother participation is never an unlock condition. Account setup/pairing is a separate unresolved onboarding choice.
BR-02	Anger/sadness may be acknowledged without labeling the child's character. Completion reflects interaction with learning material.
BR-03	New content uses supported templates and schemas. A new engine or game mechanic may require a new client release.
BR-04	Published revisions are immutable. A draft can be rejected for review without affecting the currently live revision.
BR-05	A completion record identifies learner, journey, and content revision. It remains historical after revision or withdrawal, except deliberate account erasure.
BR-06	Required digital steps must be reachable. Optional/shared/reflection/real-world steps cannot be prerequisites for required progression.
BR-07	For MVP, each child belongs to one mother account. Content admins do not inherit access to learner data.
BR-08	A device credential authorizes a specific child profile, not every child in a family.
BR-09	A withdrawn revision cannot be resumed after the client learns it was withdrawn. Historical records are retained with an unavailable-content explanation.
BR-10	Resume cursor and completion are different: revisiting an earlier scene can change the cursor but cannot undo completed steps.
BR-11	Repeating a request with the same event identifier and payload returns the same outcome; reusing that identifier with a different payload is rejected.
BR-12	Language editions share canonical step/choice IDs and completion. Translating text never resets progress or silently changes learning logic.
BR-13	A locked future preview is not a progression-locked available door. All actual unlock prerequisites must be reachable in published content.
BR-14	Worship counts, emotional improvement, and perfect consecutive days do not determine moral status or learning rewards.
BR-15	Reward and check-in writes are idempotent. Replay cannot duplicate an award; timezone changes cannot create extra daily rewards.

6. Data requirements and visibility

Conceptual entities are modeled in UML. Physical tables and a database vendor are not selected.

Data	Minimum content	Visibility / lifecycle
Mother account	Internal ID, auth-provider reference, preferences	Self and authorized account operations; no public directory.
Child profile	Internal ID, owner ID, nickname, age or age band, avatar/appearance/color/badge references	Child and owning mother within allowed views; no exact DOB/contact info required.
Pairing challenge /request	Non-plaintext verifier, expiry, pending-device reference, state	Scoped authorization only; single use, no token logging.
Device session	Session identifier, child scope, expiry/revocation metadata	Mother manages own children's sessions; tokens kept in device secure storage.
Door / journey	Stable ID, audience metadata, order, current published revision	Admin drafts private; published eligible metadata to learners.
Content revision / step / choice	Immutable revision ID, schema version, ordered step graph, assets, per-step rules	Readable to authorized learners only when available.
Attempt / checkpoint / completion	Learner, revision, cursor, completed step IDs, sequence, sync metadata	Own learner; mother sees a reduced child summary. No raw answer history stored as progress.
Reflection	Optional fixed choices; text/voice modes unresolved	No parent/admin/telemetry disclosure by default; final modality, storage, export, and erasure behavior await OQ-14.
Locale edition	Canonical revision ID, locale, edition version, localized copy/media, review state	Arabic required for release; English optional later; immutable published editions.
Practice plan / moment / check-in	Learner, approved context/practice, timezone, date, self-report	Learner-private by proposed default; mother summary excludes exact moments/check-in values.
Reward ledger / unlock	Learner, stable reward rule/event identity, value or target	Learning participation only; no raw reflection payload.
Editorial review / audit event	Reviewer/admin, revision, action, time, source/rights notes	Authorized admin operations. Not a general learner activity log.

Server progress accepts only fields needed for completion and resumption. It must not use a permissive analytics/event payload that accidentally stores child choices or reflection text. Media access must follow the same draft/published/withdrawn permissions as metadata.

7. Nonfunctional requirements - proposed targets

All numerical targets below are design proposals, not observed performance. Final test hardware, network profile, pilot concurrency, and deployment region must be recorded before baselining.

NFR-01 | Authorization

Enforce object-level ownership and role checks on every protected operation, including media preview and parent summaries. Verify with unrelated-family and wrong-role requests for each endpoint family.

NFR-02 | Sessions and pairing

Use encrypted transport and secure device credential storage. Pairing challenge expires after 10 minutes; proposed limit is 5 failed submissions per device context/IP window of 10 minutes, with additional server abuse controls. Validate atomically and rate-limit; do not rely on client checks. Exact policy is adjustable after threat review.

NFR-03 | Privacy

No ads/tracking SDKs or sensitive-answer telemetry in this MVP proposal. Review network traces and logs for pairing tokens, reflections, child contact data, and raw answers. Decide retention and account-erasure deadlines before release.

NFR-04 | Reliability

Local progress writes survive app restart once acknowledged locally. Server acknowledgment occurs only after durable commit. Retry tests shall prove no duplicate completion or backward completion regression. Explicitly label unsynced state.

NFR-05 | Interaction performance

Proposed target: local navigation/choice feedback starts within 200 ms at p95 on the agreed representative device. Media startup is measured separately. Do not block navigation on analytics.

NFR-06 | Service performance

Proposed target: metadata/progress APIs complete within 1 second at p95 at an agreed pilot load, excluding media transfer. Cold launches and large assets require their own measured budget. No capacity claim until load/device/network assumptions are agreed.

NFR-07 | Accessibility / language

RTL layouts, text/audio alternatives, captions where essential, non-color-only feedback, reduced-motion support, and accessible control names. Verify key journeys with screen readers and 200% text scaling without loss of actions. No standards-compliance certification is claimed.

NFR-08 | Compatibility

Expo React Native is fixed. OS versions, Android/iOS launch coverage, admin browser support, and device matrix remain OQ-04. A tested matrix must exist before release.

NFR-09 | Content compatibility

Unknown schemas/templates fail safely; publication validates supported client capability requirements. Test a currently supported client against a future unsupported revision without a crash or corrupt progress.

NFR-10 | Operations

Capture errors, sync failures, and publication failures with correlation IDs and no sensitive payloads. Test restore from a backup in a nonproduction environment; backup frequency, recovery objectives, retention, and operator ownership must be set before launch.

NFR-11 | Editorial quality

Every published journey has documented source/rights review and an age-appropriateness review by the project's appointed owner. Final religious text is reviewed before publication; this SRS supplies no approval.

NFR-12 | Child usability

Before release, observe children across the 10-14 range completing the key flow without coaching after setup. Record confusion, reading/audio needs, and failure to find retry/exit actions. Sample size and success threshold remain to be agreed; no untested claim that one design fits all ages.

NFR-13 | Localization

Validate complete Arabic/RTL content at launch and a nonpublic English/LTR fixture using the same canonical IDs. Test font growth, audio/caption alternatives, locale fallback, and unchanged progress after language switching. Do not require English content to launch Arabic.

NFR-14 | Practice consistency

Exercise duplicate/reordered/offline check-ins, missed days, timezone boundaries, replayed rewards, and content revisions. Records must not duplicate, reset existing completion, or turn future previews into available content.

8. External interfaces

- **Mother/child mobile app:** Expo React Native. Specific screens are design work to follow, not mandated pixel layouts.
- **Admin client:** proposed desktop web app. Writes through authenticated content operations, not direct database access.
- **Identity boundary:** authenticate adults, issue scoped child sessions, revoke credentials, and recover adult access. Provider and login method remain open.
- **Content interface:** paginated/ordered catalog, revision manifest, capability metadata, and authorized media references.
- **Progress interface:** scoped attempt creation, version-bound checkpoint events with idempotency keys, and confirmed-summary reads.
- **Media pipeline:** uploads, validation/processing status, published-delivery references, and withdrawal checks. Provider, codecs, limits, and signed-URL policy are implementation decisions.

No API route names, database engine, hosting platform, or microservice topology is contracted by this document.

9. Failure and change policy

- Expired pairing: regenerate; never link by nickname or accept a guessed identity.
- Interrupted learning: resume locally when required cached data exists; otherwise preserve checkpoint and request connectivity.
- Routine content update: keep current attempt on its version; new attempts use the new version.
- Content withdrawal: stop subsequent authorized delivery; keep an explanatory historical record; do not claim cached bytes can be remotely erased instantly.
- Stale progress submission: merge completion monotonically, validate cursor ordering, and return authoritative state. Device replacement starts from server state.
- Failed publish: keep prior live version and return actionable validation errors.
- Account erasure: stop access immediately; apply the agreed erasure/backup policy without promising a period not yet defined.

10. Release gate and outstanding decisions

Use Section 08 for acceptance scenarios. Before a public release, resolve account setup/recovery, mother visibility, launch platforms/countries, content ownership, data retention/erasure, operational ownership, and the test/device/load matrix. Content production and review are release dependencies, not assumed completed work.

05 Detailed use cases

UC-01 - Establish family access on separate devices

Actors: Mother, child. **Requirements:** FR-01-FR-05. **Assumptions:** P-01/P-02/P-03; initial adult setup is acceptable, pending OQ-09.

Preconditions: Mother can authenticate; service is reachable. A device does not gain child access merely by choosing “child” on a welcome screen.

Main flow:

1. Mother signs in, reads the proposed profile/linking explanation, and creates a child profile.
2. Service binds the profile to her account; she requests a pairing challenge.
3. Child enters that challenge on their separate device.
4. Service checks validity and reserves a single pending redemption atomically. It does not yet issue a child session.
5. Mother sees the pending request and confirms the device she is pairing.
6. Service checks ownership and challenge expiry again, atomically consumes the challenge, and issues a child-scoped session to the pending device through its protected pending-request credential.
7. Child device stores the session securely and loads its own published catalog and progress.

Alternatives/failures: invalid/expired challenge to retry or regenerate; wrong device to reject/cancel and regenerate; simultaneous redemptions to only one pending reservation; mother does not confirm before expiry to no child session; interrupted completion response to the same pending device can retry and retrieve the same authorization result, not mint an additional session.

Postcondition: authenticated child device is linked to exactly one profile. Mother credentials are not copied to it. No learning step requires further mother confirmation.

Unresolved: adult login/recovery provider, explicit notices, final abuse policy, whether a guest preview should exist.

UC-02 - Explore the first child journey

Actor: Child. **Requirements:** FR-06-FR-11. **Preconditions:** authorized session, published supported content available.

1. Child completes character setup, enters «خريطة الحياة الطيبة», and opens the available «باب الذكر» entry experience.
2. Service selects an available revision and creates/resumes a bound attempt.
3. Child views an illustrated scene; narration is optional and core meaning is readable.
4. Child explores a choice and its authored consequence.
5. Child may try a different response and continues after the step's participation criterion is satisfied.
6. Child sees a real-life activity and may defer it; shared participation and reflection are optional.

Alternatives/failures: unsupported schema to explanatory state with safe return; missing required media to retry without losing checkpoint; optional media failure to text alternative; distressing/repeated-mistreatment scenario to authored trusted-adult guidance, not a diagnostic or emergency service.

Postcondition: the child has a meaningful next action and saved learning checkpoint. No moral score is generated.

Dhikr content is provisional: a notification interrupts study; the child explores a reconnecting choice, discovers the meaning of Dhikr through a story, and tries a brief guided practice. The app accepts “not much changed” without grading emotions. Final Arabic copy, religious references, pacing, and assets require review.

UC-03 - Save, resume, and complete learning

Actors: Child or mother learner. **Requirements:** FR-12, FR-13, FR-22, FR-25, FR-26.

1. App records the current checkpoint and completion evidence locally using a unique event ID and attempt revision.
2. App marks unsynced progress as pending and submits allowed progress fields to the service.
3. Service checks learner scope, revision availability, step identity, event uniqueness, and sequencing.
4. Service atomically records the event and merges completed steps without regression.
5. Service determines completion from the bound revision's required-step rules and returns confirmed state.
6. App acknowledges synchronization and can resume at the saved checkpoint on later entry.

Alternatives/failures: duplicate request to same result; same event ID/different payload to reject; connection loss to durable queue; force-close to restore local checkpoint; stale cursor event to retain newer cursor; a deliberate later replay of an earlier scene to allow newer cursor while retaining completion; withdrawn revision to stop new content access, retain historical records, show explanation.

Postcondition: progress is either explicitly pending locally or committed on the server. A second device cannot see progress that has never reached the server.

Privacy: choice feedback is processed from authored content; raw choice histories and optional reflection answers are not progress payloads. Completion evidence contains step IDs/statuses, not free-form answers. The server is not certifying real-world behavior.

UC-04 - Complete the mother's personal learning

Actor: Mother. **Requirements:** FR-05, FR-14.

Preconditions: authenticated mother, reviewed published mother content.

1. Mother opens her personal journey, separately from the child overview.
2. She reads/listens to the lesson, considers the suggested personal practice, and completes its required digital steps.
3. Her progress is saved using her own learner identity.
4. Optional reflection can be skipped. Text/voice modalities and storage remain subject to OQ-14; reflection payloads are separate from progress.

Alternative: child has not started, has completed, or is unavailable to mother's learning is still accessible.

Postcondition: mother completion is independent of every child's attempt.

UC-05 - Support a child optionally

Actor: Mother; child participates only if desired. **Requirements:** FR-10, FR-15, FR-16.

1. Mother chooses one of her linked children.
2. Service returns topics and server-confirmed completion only, through a summary projection.
3. Mother opens an authored support suggestion or shared activity.
4. They may discuss it outside the app; no simultaneous device connection or chat is required.

Alternatives: no child progress yet to helpful empty state; child progress pending on another device to last confirmed state, not invented freshness; activity declined to child can use the solo route and keep progressing.

Postcondition: participation does not unlock or block a required child step. Mother has not obtained hidden access to answers/reflections.

UC-06 - Author, review, and publish new content

Actor: Admin. **Requirements:** FR-17-FR-21, FR-24.

Preconditions: admin authorization, supported templates known; no learner-data access implied.

1. Admin creates a door/journey or a draft revision of an existing journey.
2. Admin assigns child/mother/shared paths, stable step IDs, ordering, template data, media, and completion rules.
3. Media is uploaded privately; processing and rights/accessibility metadata are checked.
4. Admin previews audience-specific paths and records editorial review, including sources where applicable.
5. Validator checks schema, assets, references, reachable required steps, optionality rules, and supported client capabilities.
6. Admin explicitly publishes using the expected current draft version.
7. Service atomically freezes the revision, changes the live pointer, and records an audit event.
8. Existing compatible clients see the new journey on the next catalog refresh; already open sessions are not forcibly navigated away.

Alternatives/failures: invalid data/unreviewed material to actionable errors; stale draft version to reload/reconcile; publication failure to previous live version remains; upload alone to remains private; same publish operation retried to one published revision and one effective publication action.

Postcondition: a complete revision is live. No app update is needed for content using already supported templates; no arbitrary game-engine creation is implied.

UC-07 - Revise or withdraw content

Actor: Admin. **Requirements:** FR-21-FR-24.

Routine revision: clone live revision to a new draft to review/validate to publish. Existing attempts remain bound to the older available revision. New attempts use the current one. Old completion remains historical.

Withdrawal: admin chooses an exact revision to provides reason to confirms scope to service marks it withdrawn and invalidates new delivery access. Bound attempts receive an unavailable state on their next authorization/content check. Their records are not erased.

Alternatives: no replacement available to honest unavailable state, no fake next journey; correction to withdrawn content to new reviewed revision; previous safe version desired to create a new draft from it and review/publish, not an unreviewed silent rollback.

Limit: already downloaded bytes cannot be guaranteed to disappear immediately from disconnected devices. The offline policy remains under review.

UC-08 - Revoke device access or erase a profile

Actor: Mother. **Requirements:** FR-04, FR-27.

Revoke/replace: mother selects the relevant device to confirms revocation to service invalidates it to protected requests fail to replacement device is paired using UC-01 and restores server-confirmed progress.

Erase child profile: mother sees what will be removed to explicitly confirms to sessions are revoked to profile and learner records enter the agreed deletion process to local clients clear scoped caches on receiving revocation/deletion or logout.

Erase mother account: explain inclusion of all owned child profiles under P-03; require confirmation; revoke all affected sessions and schedule linked-data removal.

Failure: unrelated-account request to reject without revealing profile details. Backup purge and retention timing are unresolved release decisions; do not promise immediate erasure from every backup.

Postcondition: a lost device cannot make new authorized server requests. Revocation is not equivalent to deletion; historical progress survives revocation but is removed/anonymized according to confirmed erasure policy when deletion is requested.

UC-09 - Plan and continue Dhikr practice

Actor: Child. **Requirements:** FR-31-FR-34. **Status:** proposed interpretation of the supplied Dhikr concept.

1. Child discovers daily remembering moments, such as before sleep or before studying.
2. Child chooses one approved context and practice for a when/then plan.
3. A suggested seven-day practice window begins in the plan's chosen timezone.
4. Child may check in with tried/remembered/remembered independently, or leave a day blank.
5. App acknowledges participation, resumes after pauses, and offers the closing reflection without claiming that a habit or calmer emotion has been proven.

Failures/alternatives: no check-in does not reset learning; retry writes once; changing devices uses confirmed server state; overlapping plans do not multiply the same reward. Any XP rules are explicit, capped by eligible event identity, and unrelated to worship counts.

Postcondition: saved practice plan and voluntary history; digital completion and long-term practice remain distinct. Compulsory seven-day gating is unresolved.

UC-10 - Create a character and navigate doors

Actor: Child. **Requirements:** FR-29, FR-30.

1. After linked access, child chooses avatar, nickname, appearance, color, and badge from the proposed preset catalog.
2. Choices are saved; the child enters «خريطة الحياة الطيبة».
3. Available Dhikr entry opens its introductory discovery; later authored regions explain their learning prerequisites.
4. Selecting a future preview such as «معية الله» explains that the door is not yet available, without exposing draft content or asking for impossible unlock tasks.

Alternatives: canceled customization retains last saved settings; missing asset has a valid default; withdrawn content takes precedence over prior unlock state. No parent approval is required for each learning unlock.

UC-11 - Publish a language edition without resetting progress

Actor: Admin; learner is the eventual reader. **Requirements:** FR-28, FR-35, NFR-13.

1. Admin opens an existing canonical content revision and selects Arabic or a later English edition.
2. Admin supplies localized labels, scene/choice/feedback text, audio/captions, and accessibility copy linked to stable IDs.
3. Validation checks complete required fields, media readiness, religious translation review, and locale-specific rendering.
4. Publishing an immutable locale edition makes it eligible for the catalog's enabled-language policy.
5. The learner sees the appropriate published edition with the same progress and unlock identity.

Alternatives: English may stay absent or draft while Arabic launches; missing English falls back explicitly to Arabic after English is enabled; no mixed-language sentence fallback; changed learning meaning/graph requires a new canonical revision rather than a translation-only edit.

UC-12 - Optional closing reflection

Actor: Child. **Requirements:** FR-26. **Decision dependency:** OQ-14.

The closing prompts ask when Dhikr felt easiest, when it was difficult, and where the child would like it in their day. The source proposes writing or a voice note. Those input modes are recorded as requests, not silently enabled.

The child may skip without losing learning progress. Any selected implementation must define storage, retention, deletion, and microphone denial/interruption behavior before release. Raw answers remain excluded from parent summaries, content-admin tools, ordinary progress synchronization, and analytics under the current privacy proposal. No AI analysis or voice transcription is implied.

06 Logical architecture

1. Direction

Start with one backend application organized into clear modules, a transactional persistence boundary, and separate media storage/delivery. These are logical responsibilities, not a recommendation to deploy separate microservices. The admin web client is a proposal. Backend framework, identity provider, database, storage/CDN, queues, hosting, and observability products remain undecided.

The content model is the key architectural choice: administrators publish data that existing client renderers understand. This supports new journeys without turning the MVP into a general game-creation platform.

2. Responsibility boundaries

Component	Owns	Does not own
Mobile role shell	Mother/child navigation and accessibility	Server authorization decisions.
Content renderer	Supported template display, authored choices, audio/text controls	Executing remote scripts or inventing new game mechanics.
Local progress store and outbox	Durable checkpoints, pending events, restart recovery	Claiming server sync before acknowledgment.
Identity/family module	Adult identity mapping, ownership, pairing, child sessions, revocation/deletion initiation	Editorial publication.
Catalog/content module	Drafts, immutable revisions, validation, availability, publication pointer	Treating a new schema as supported by old apps.
Progress module	Version-bound attempts, idempotent events, required-step completion	Moral scores or raw reflection storage.
Mother-summary projection	Allowed child topic/completion summary	Pass-through access to detailed learner event data.
Media module	Private upload, type validation, asset readiness, delivery authorization	Automatic publication on upload.
Admin client	Editing, review, preview, explicit publication	Direct database writes or learner-data browsing.
Audit/operations	Content lifecycle audit, non-sensitive faults, backup/restore	Child-answer analytics.

3. Proposed trust boundaries

All clients are untrusted. Server operations derive ownership from validated identity/session scope; passing a child ID never grants access. Preview media is private to authorized admins. Published media is available only according to its revision's delivery policy. Catalog filtering is not sufficient access control for direct asset or revision requests.

A pairing challenge is not itself a permanent credential. A pending device must possess its own pending-request credential, and the owning mother must confirm before issuance. Pending requests expire; approval, redemption,

and consumption must be atomic and duplicate-safe. The precise cryptographic protocol belongs in the detailed auth design, not this conceptual UML.

Content admins manage content only. A future support capability requires an explicit permission and purpose; it must not be silently added to “admin”.

4. Content contract

Each revision has a schema version and declares required client capabilities. Each step has a stable ID within that revision, audience path, template type, authored data, optional media references, and a completion rule. Choice destinations stay inside their authored path unless an explicit supported transition exists.

Publish validation checks:

- Assets exist, are ready, and have the required alternatives/source/rights metadata.
- Required digital steps are reachable; optional tasks do not become hidden gates.
- All references and choice targets resolve; authored loops have an exit.
- Template/schema capabilities are compatible with the intended supported clients.
- Review is recorded and the expected draft version matches the saved version.

Published data is immutable. The active catalog points to a live revision; previous nonwithdrawn revisions remain available for bound attempts. Withdrawal changes availability, not historical content or completion. A content fix requires another reviewed revision.

5. Progress and synchronization

Use distinct learner identities for mother and child. An attempt binds one learner to one content revision. Persist a resume cursor separately from the set of completed required steps.

Proposed event envelope: eventId, attemptId, revisionId, deviceSessionId, sequence, checkpointStepId, and newly completed allowed step IDs. Exclude raw answer/reflection payloads. The server validates step IDs and state transitions; completion measures participation only, so clients are not proving real-world conduct.

Local storage commits before showing “saved on this device”. The outbox retries with the same identifier. The server atomically deduplicates and commits before returning “synced”. Same identifier with a changed payload is an error. Stale cursor updates cannot overwrite a newer cursor; completed steps merge monotonically.

The draft assumes one actively used child device session per child for learning; replacement revokes the previous learning session. This reduces concurrent-cursor ambiguity. If multiple simultaneous child devices are required, add an explicit conflict policy before implementation. Multiple children on their own devices remain supported under P-03.

Full offline playback is not promised. Cached scenes may be usable during a disruption; missing media requires connectivity. A replacement device only restores server-confirmed state. Optional reflection data is separate from the progress outbox. Written/voice storage and deletion design remain conditional on OQ-14; no existing synchronization endpoint should accept those payloads by default.

6. Publication consistency

A publish operation uses an idempotency identifier plus expected draft version. Validate before mutation; commit immutable revision, live pointer, and audit record in one transactional boundary. External media must be ready before the transaction. A failure leaves the prior live pointer unchanged. Asset cleanup after abandoned drafts needs a later retention policy.

For withdrawal, deny new authorized delivery and invalidate cached metadata where feasible. Time-bounded media links can reduce exposure, but exact link lifetime/CDN policy is undecided. Do not claim that previously delivered files can be retracted instantly.

7. Data and API design still to come

After the behavioral baseline is accepted, produce:

- Physical schema with keys, constraints, indexes, deletion/retention behavior, and migrations.
- API contracts for accounts/pairing, catalog/revisions, progress, summaries, and admin workflows.
- Concrete auth threat model, credential lifetime/rotation/recovery, abuse controls, and ownership tests.
- Media format/size limits, processing pipeline, caching and storage budget.
- Deployment topology, environment separation, CI/CD, backups, restore targets, monitoring, and cost estimates.

These are not omitted implementation work: the current stage establishes requirements and UML before committing to a stack. Picking providers before OQ-04/OQ-05/OQ-06/OQ-10 would conceal assumptions about platform, connectivity, distribution, and scale.

8. Main risks and proposed controls

Risk	Proposed control	Decision still needed
Pairing the wrong device	Expiry, single pending redemption, mother confirmation, revocation	Accept onboarding mechanism.
Leaking one family's records	Server ownership checks and explicit response projections	Confirm summary visibility.
A content edit breaks an ongoing journey	Immutable revisions and version-bound attempts	Withdrawal/replacement UX.
Admin publishes unsupported interactions	Capability validation and bounded templates	Final template list.
User assumes unsynced work is backed up	Visible local/pending/confirmed state	Connectivity scope.
Production content is missing or unsuitable	Editorial ownership, provenance, preview, age testing	Content team and review process.
Broad scope overwhelms single-door MVP	One complete vertical journey plus reusable authoring	Approve the proposed MVP boundary.

9. Localization without duplicating the curriculum

Keep locale-neutral Door, Journey, Stage, Step, and Choice identities. A versioned **LocaleEdition** belongs to one canonical ContentRevision and maps stable content keys to text, audio, captions, and accessibility alternatives. Localized door/journey labels are catalog metadata. Avoid titleAr/titleEn columns and duplicated English journeys as the long-term structure.

Arabic is complete and enabled at launch. English editions can be added and reviewed later without changing progress identity or the underlying content graph. Each published edition is immutable; a correction creates another

locale-edition version. A semantic change to the learning graph needs a new canonical revision. Attempts can pin an edition during a session and adopt a reviewed equivalent edition on next entry without losing progress.

Fallback is explicit: Arabic remains the baseline; incomplete English is not published and is not mixed into Arabic UI. The mobile shell still needs translated interface strings, LTR layout, and regression testing for an eventual English release; content readiness alone does not ship a bilingual app.

10. Practice, rewards, and map availability

Add logical modules for character configuration, practice plans/check-ins, participation rewards, and map eligibility within the same proposed backend. Store approved avatar-option IDs; personal media uploads are not a prerequisite.

Represent future_preview, progression_locked, available, completed, and withdrawn separately. Preview-door metadata is explicitly published; unpublished educational assets remain private. Server eligibility checks use reachable learning prerequisites and content availability. Previously unlocked content can still be withdrawn.

PracticePlan identifies learner, approved moment/practice, timezone, and optional target window. PracticeCheckIn is unique by plan and local date; updates are idempotent, with no punitive missing-day reset. A proposed RewardGrant uses learner plus stable eligible reward/event identity to avoid replay farming. These records never certify worship or emotional improvement. The mother's summary excludes detailed moments/check-in values unless that policy is explicitly changed.

The source's seven days, ten-second exercise, ten-minute introduction, XP values, named characters, garden, and secret doors are authoring proposals. A simple governed content implementation does not imply background reminders, AI evaluation, sound recording, or arbitrary mini-game generation. Resolve reflection input/storage and practice completion gates before implementing those dependent behaviors.

07 UML models

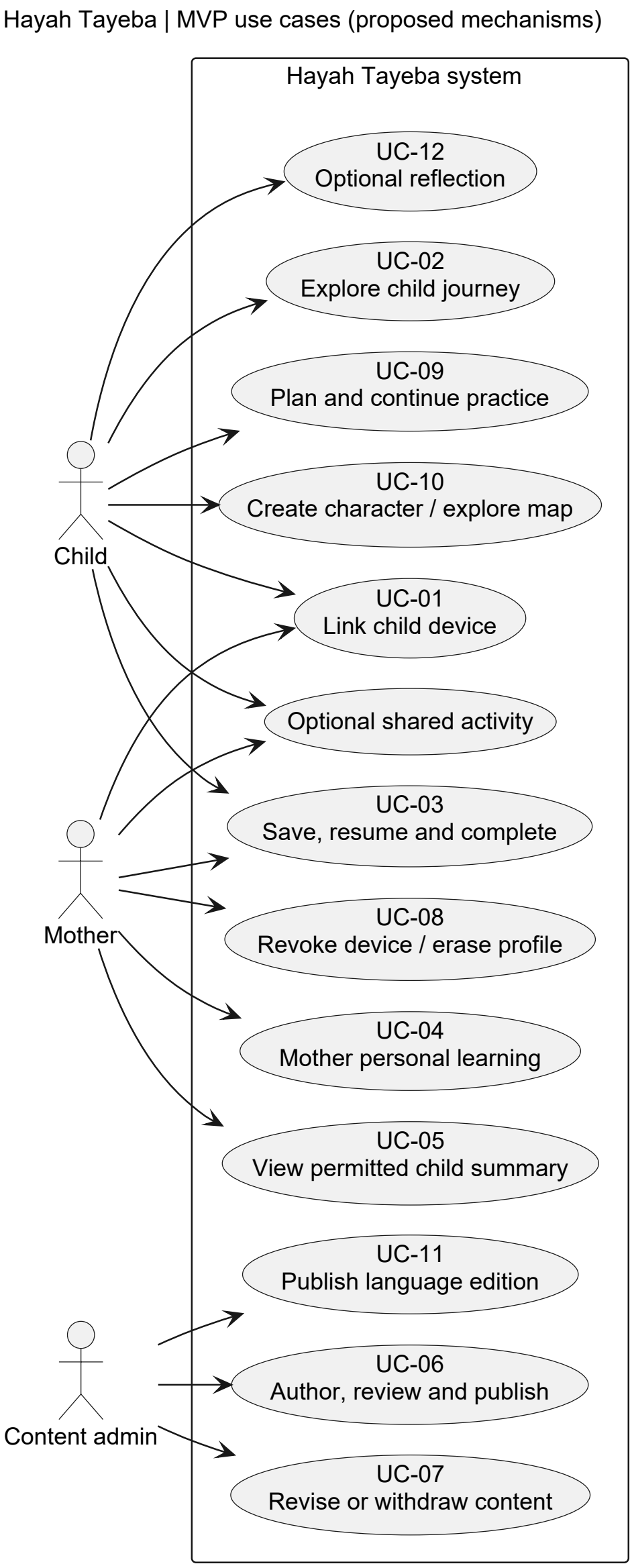
The following ten models express the draft requirements. Behavioral mechanisms, privacy boundaries, and account ownership marked as proposals remain subject to review. These are software models, not UI layouts.

Model	Purpose
Use cases	Three human actors and the system goals. Initial linking involves the mother; learning participation does not depend on her availability. Content administration grants no access to learner records.
Content domain	Conceptual classes, not physical database tables. Stable journeys have immutable revisions with audience-specific paths. Multiplicities describe a valid publishable model; drafts may be incomplete.
Learner domain	Mother and child are separate learners. Attempts bind to a content revision. The proposed ownership and active-device restrictions remain review decisions.
Child learning activity	Independent learning, durable local progress, optional shared/reflection steps, and network recovery. A locally finished session can still have synchronization pending.
Device pairing sequence	Proposed temporary-code flow with explicit mother confirmation before granting child-scoped device access. Detailed credentials and identity-provider selection remain open.
Progress synchronization sequence	Separates local save from server confirmation and handles duplicate submissions. The mother receives an authorized summary rather than raw choices or reflections.
Content revision lifecycle	Publication freezes a revision. Editing creates another draft; withdrawal affects future availability without silently removing historical completion.
Logical components	Proposed client, backend, persistence, identity, and media boundaries. Backend modules do not imply separately deployed microservices or a selected vendor.
Localization model	Arabic release and future English editions share canonical content IDs and progress. Localized text and media are versioned, reviewed editions.
Practice and rewards	Remembering moments, plans, voluntary check-ins, and participation rewards. Reflection input/storage and seven-day gating remain open.

Drawing pages use A3 dimensions so the original diagrams remain legible when zoomed or printed at their native size. Editable PlantUML sources are maintained with the project specification.

UML 01 Use cases

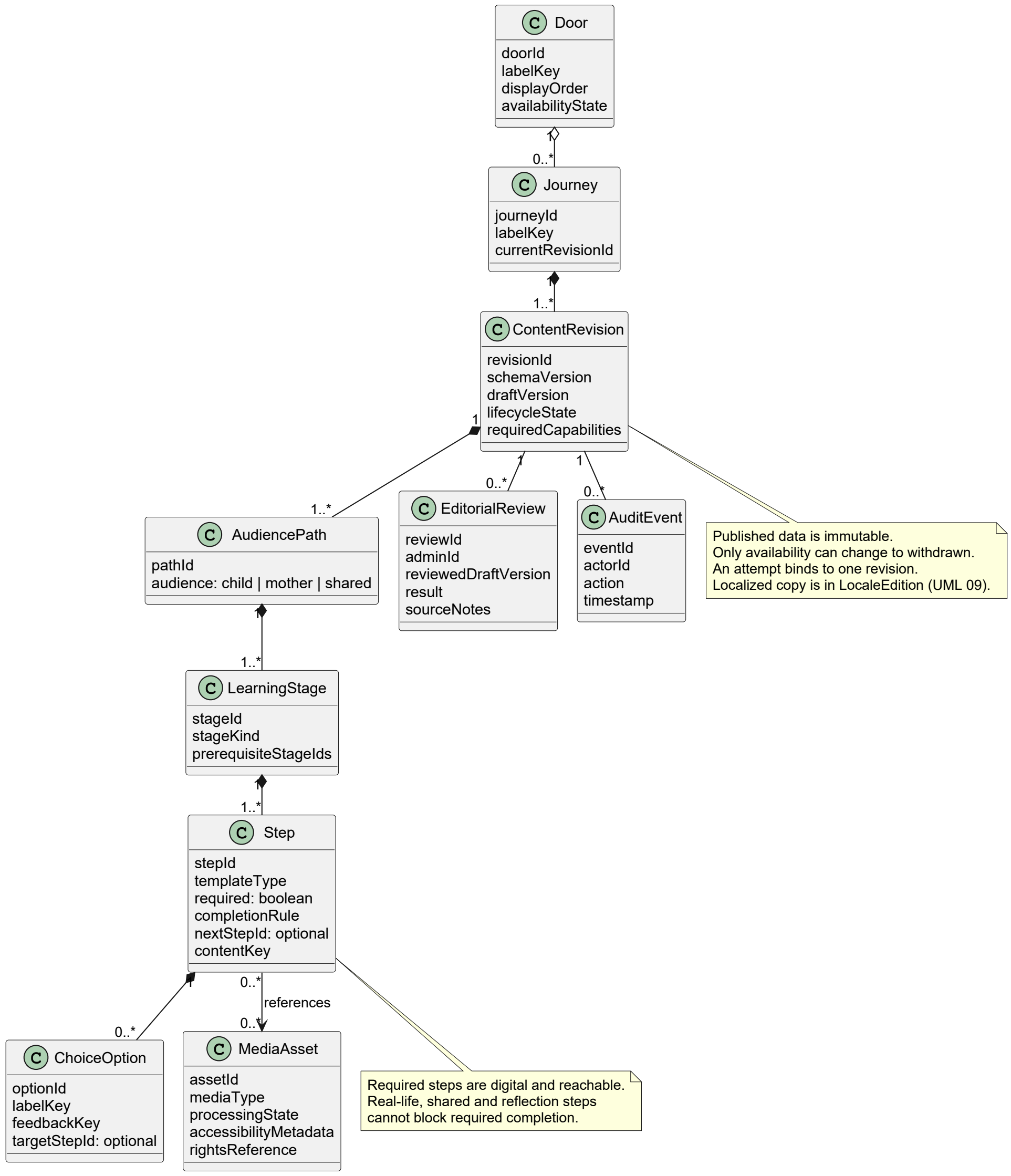
Three human actors and the system goals. Initial linking involves the mother; learning participation does not depend on her availability. Content administration grants no access to learner records.



UML 02 Content domain

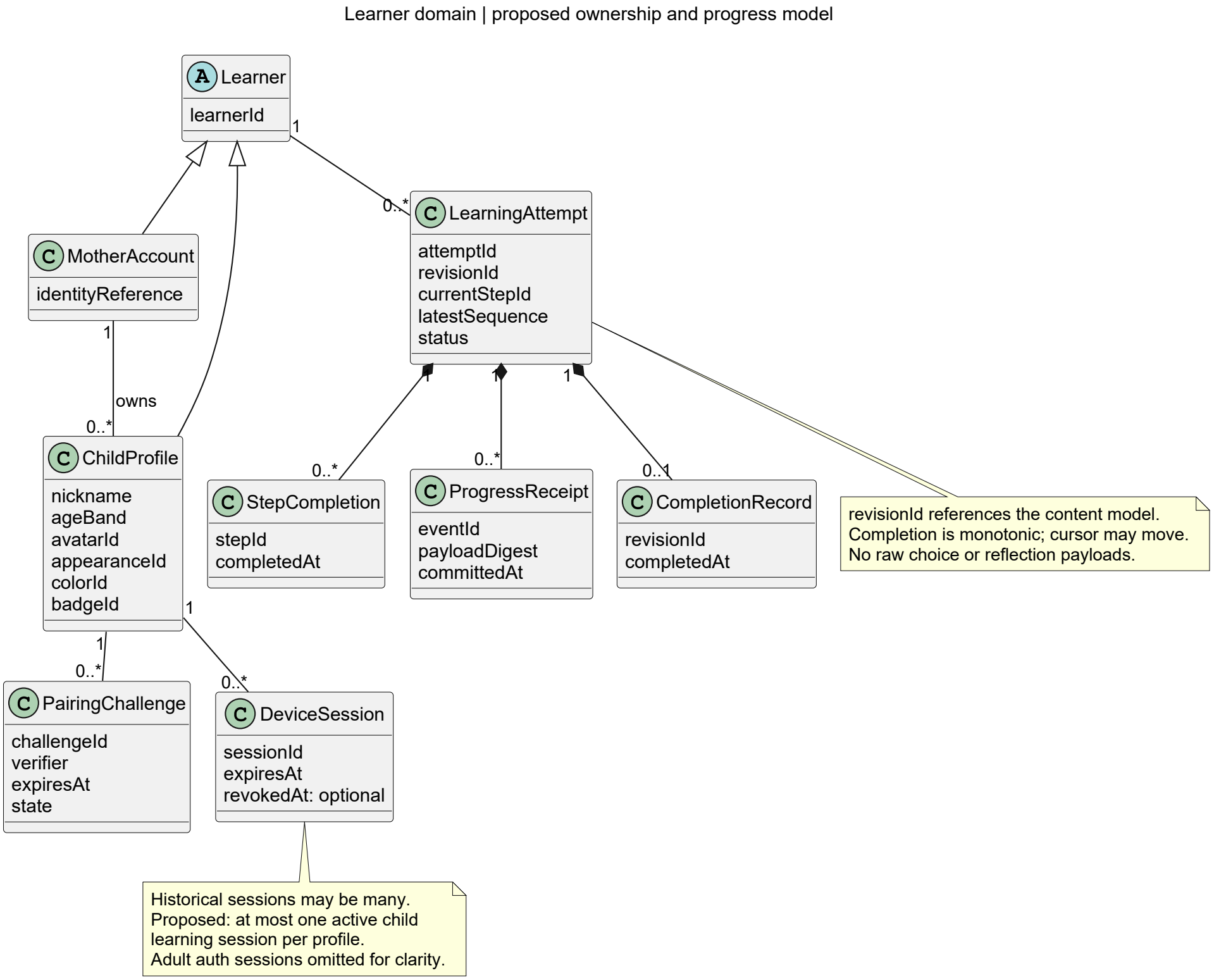
Conceptual classes, not physical database tables. Stable journeys have immutable revisions with audience-specific paths. Multiplicities describe a valid publishable model; drafts may be incomplete.

Content domain | conceptual classes, not database tables



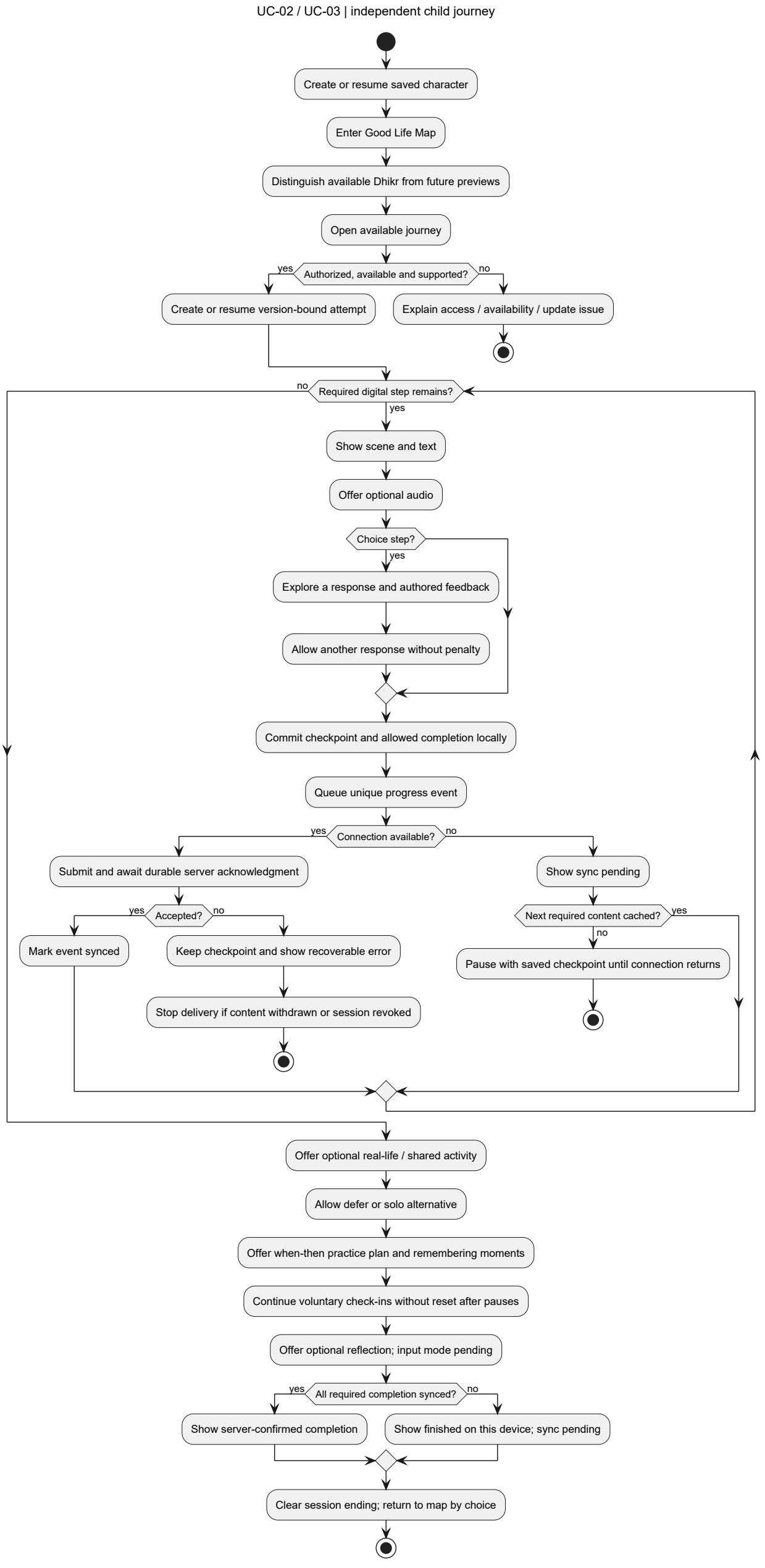
UML 03 Learner domain

Mother and child are separate learners. Attempts bind to a content revision. The proposed ownership and active-device restrictions remain review decisions.



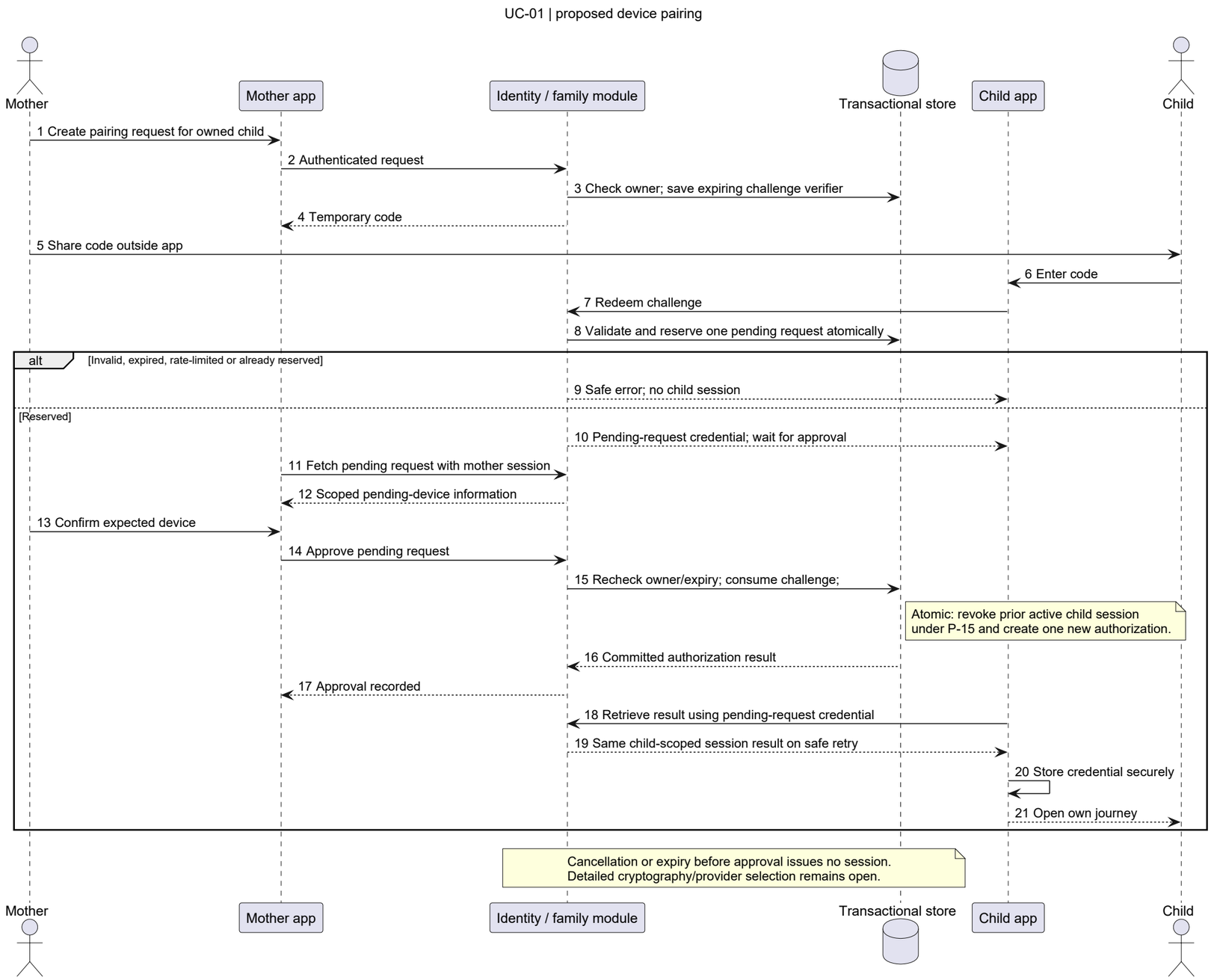
UML 04 Child learning activity

Independent learning, durable local progress, optional shared/reflection steps, and network recovery. A locally finished session can still have synchronization pending.



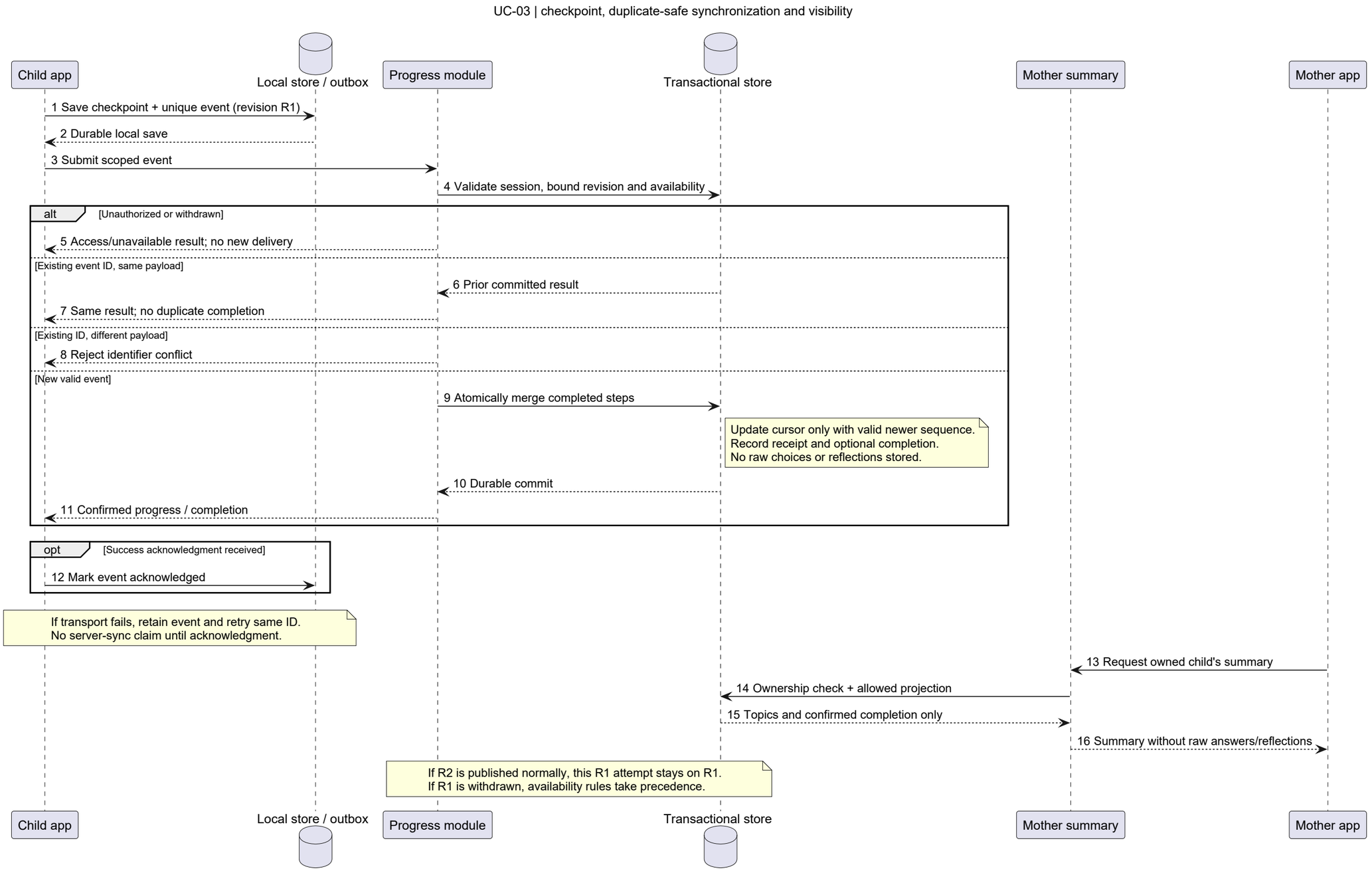
UML 05 Device pairing sequence

Proposed temporary-code flow with explicit mother confirmation before granting child-scoped device access. Detailed credentials and identity-provider selection remain open.



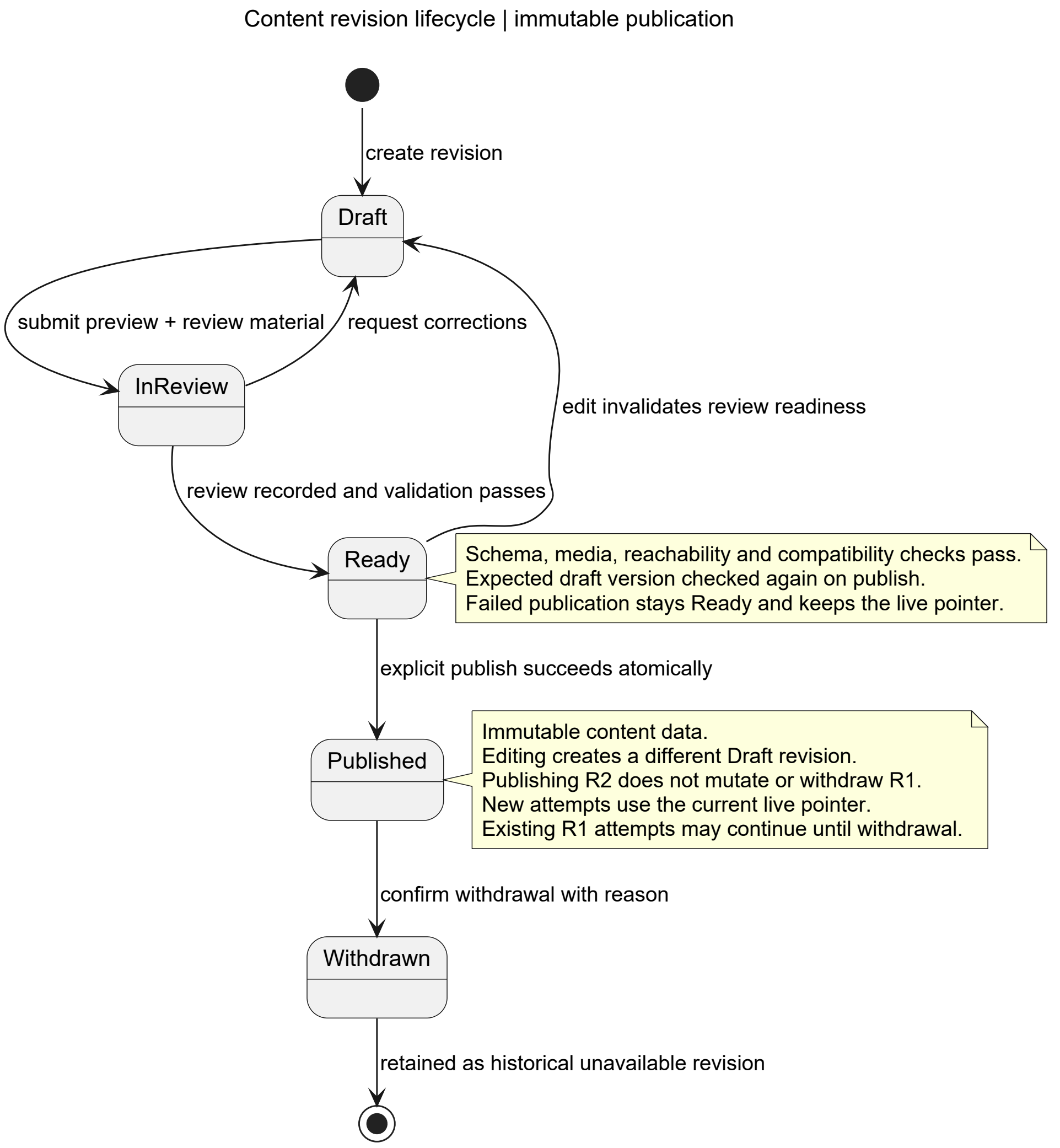
UML 06 Progress synchronization sequence

Separates local save from server confirmation and handles duplicate submissions. The mother receives an authorized summary rather than raw choices or reflections.



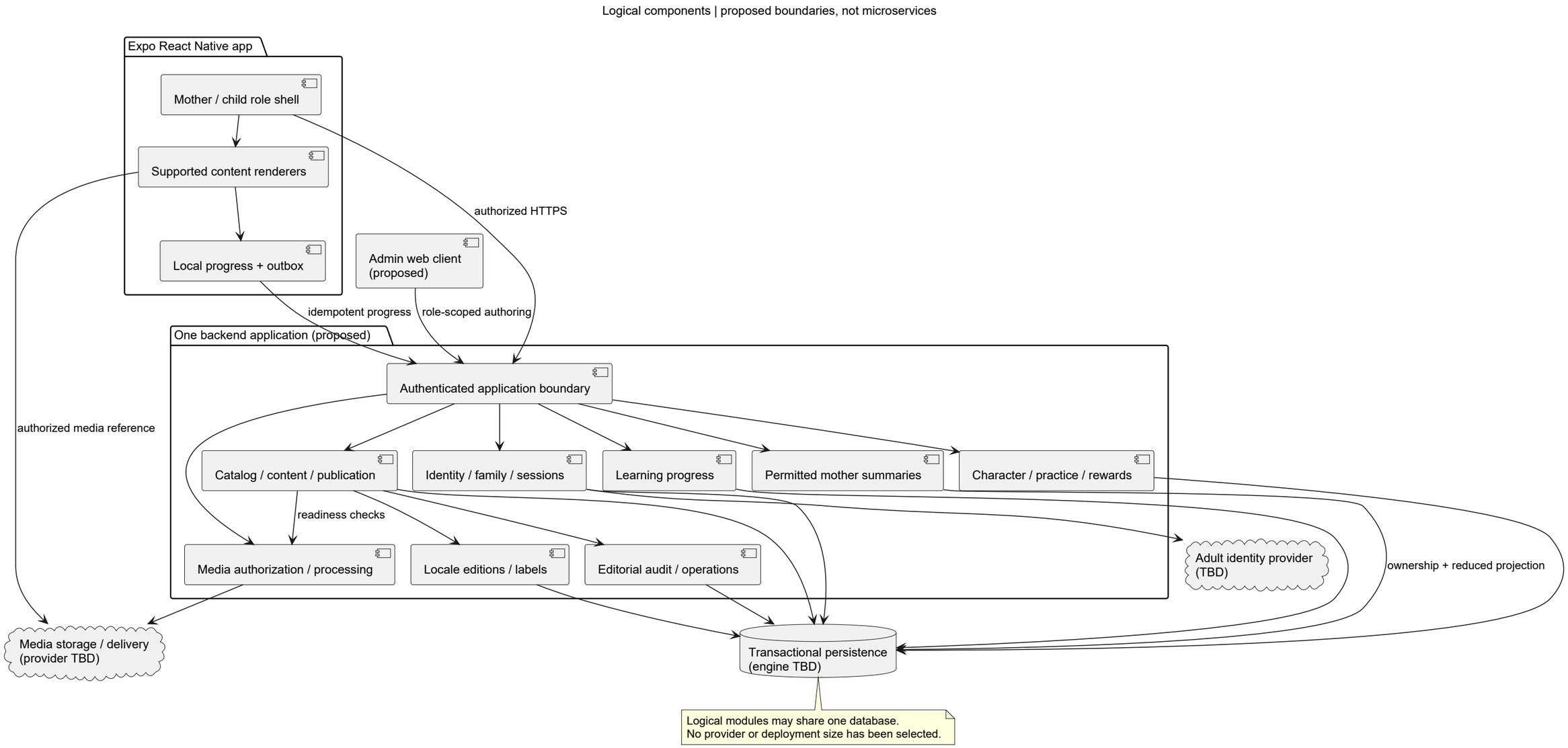
UML 07 Content revision lifecycle

Publication freezes a revision. Editing creates another draft; withdrawal affects future availability without silently removing historical completion.



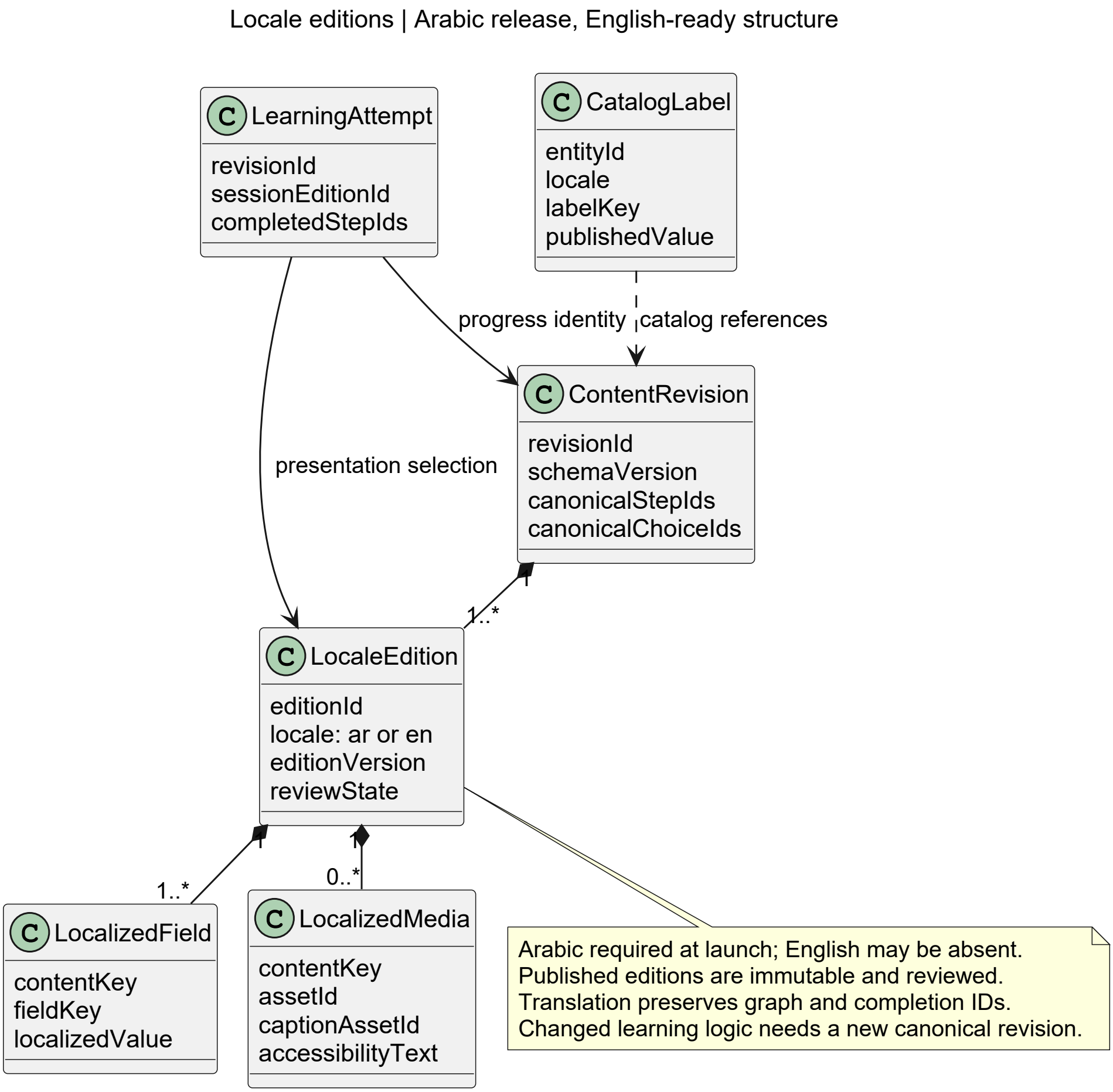
UML 08 Logical components

Proposed client, backend, persistence, identity, and media boundaries. Backend modules do not imply separately deployed microservices or a selected vendor.



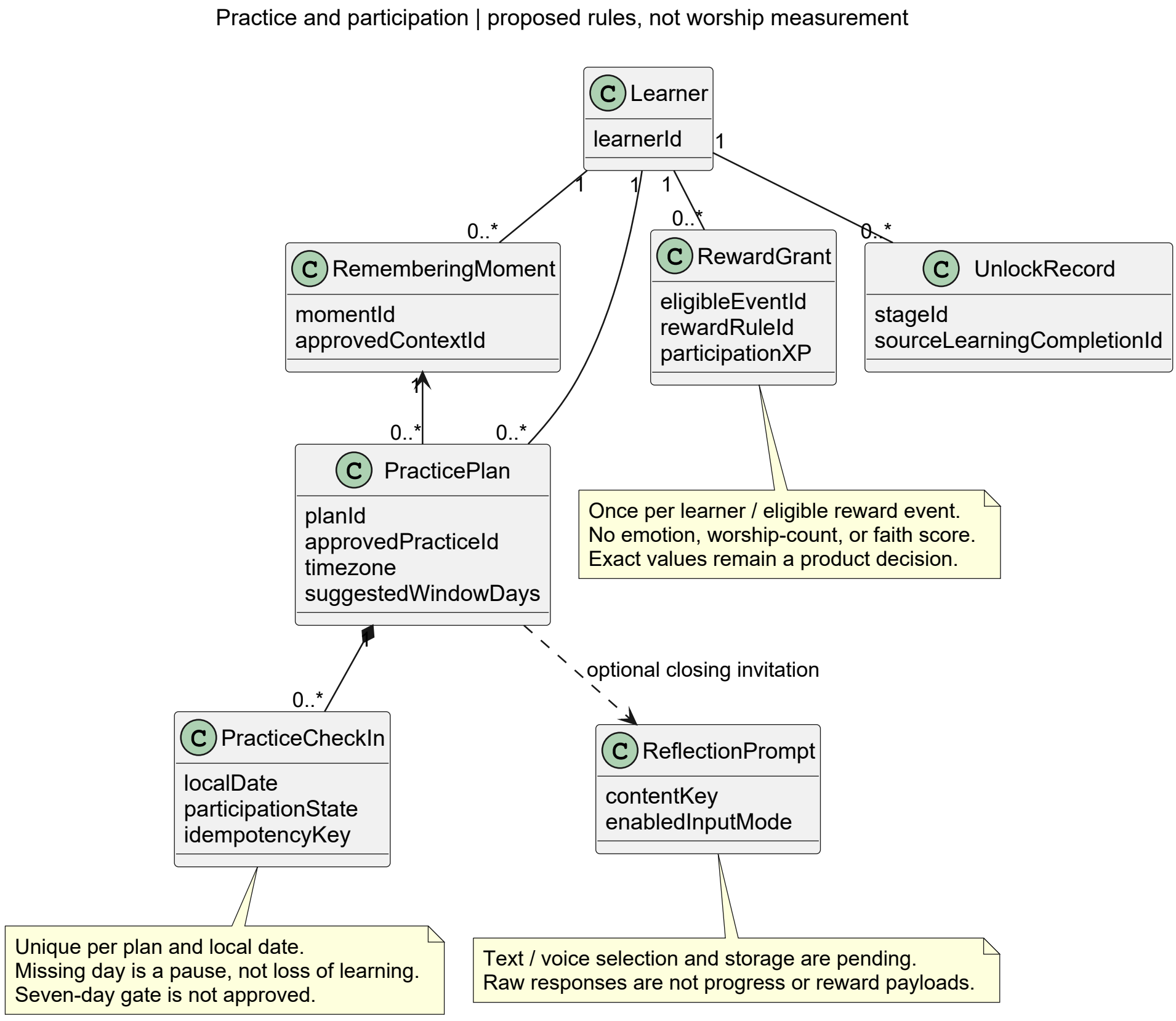
UML 09 Localization model

Arabic release and future English editions share canonical content IDs and progress. Localized text and media are versioned, reviewed editions.



UML 10 Practice and rewards

Remembering moments, plans, voluntary check-ins, and participation rewards. Reflection input/storage and seven-day gating remain open.



08 Requirements traceability

These are planned acceptance checks. They have not been executed against an application.

Model keys: U1 use cases; U2 content classes; U3 learner classes; U4 child activity; U5 pairing sequence; U6 progress sequence; U7 content lifecycle; U8 components; U9 locale editions; U10 practice and rewards. See the numbered models in Section 07.

Functional coverage

Requirement	Use case	Models	Planned acceptance check
FR-01	UC-01, UC-06	U1, U5, U8	AT-01: try anonymous, wrong-role, and unrelated-family requests; deny each protected operation.
FR-02	UC-01	U3, U5	AT-02: create a valid child profile without contact data; reject unsupported age and cross-owner binding.
FR-03	UC-01	U5	AT-03: happy-path pairing; expired/cancelled/reused code; concurrent redemption; wrong-owner approval; retry lost issuance response.
FR-04	UC-08	U3, U5	AT-04: revoke old device; reject its next protected request; replacement restores only server-confirmed progress.
FR-05	UC-01, UC-04	U3, U8	AT-05: mother and child completion remain separate; child cannot enter mother's account context.
FR-06	UC-02	U1, U4	AT-06: one full door and explicit preview metadata work; draft and withdrawn lessons cannot start.
FR-07	UC-02	U2, U4, U8	AT-07: render supported manifest; safely reject future schema/template without corrupting an existing attempt.
FR-08	UC-02	U2, U4	AT-08: read/listen/navigation with audio off, unavailable, or interrupted; essential meaning remains accessible.
FR-09	UC-02	U2, U4	AT-09: every authored choice resolves; try another response; no moral score or completion reset.
FR-10	UC-02, UC-05	U1, U4	AT-10: complete all required learning with no mother participation after setup.
FR-11	UC-02	U2, U4	AT-11: defer real-life task and skip reflection; required progression still works.
FR-12	UC-03	U3, U6	AT-12: save, kill app, resume, lose network, retry duplicate event, reject changed-payload ID reuse.
FR-13	UC-03	U3, U4, U6	AT-13: required-step set yields one completion per attempt; optional skips/replay preserve history.
FR-14	UC-04	U1, U2, U3	AT-14: mother starts and completes her own reviewed content regardless of child state.

Requirement	Use case	Models	Planned acceptance check
FR-15	UC-05	U6, U8	AT-15: inspect parent-summary response; only linked child's permitted fields appear.
FR-16	UC-05	U1, U4	AT-16: shared activity can be declined or done asynchronously; solo route remains available.
FR-17	UC-06	U1, U2, U8	AT-17: publish a second supported journey; unchanged client binary retrieves it.
FR-18	UC-06	U2	AT-18: validate every supported template, malformed step, broken target, and optional-step gate.
FR-19	UC-06	U2, U8	AT-19: incomplete/invalid upload cannot publish; draft media is inaccessible to child session.
FR-20	UC-06	U2, U7	AT-20: preview paths and require review; fail on missing alternatives, bad sources/references, unreachable steps.
FR-21	UC-06, UC-07	U2, U7	AT-21: failed, repeated, or stale publish leaves an atomic live pointer and no partial revision.
FR-22	UC-03, UC-07	U2, U3, U6	AT-22: publish R2 during an R1 attempt; old attempt stays on R1 and historical completion survives.
FR-23	UC-07	U7, U8	AT-23: withdraw bound revision; block subsequent online delivery, preserve records, show unavailable state.
FR-24	UC-06, UC-07	U2, U8	AT-24: verify actor/version/action/outcome audit records without credentials or child answers.
FR-25	UC-02, UC-03	U4, U6	AT-25: fail required/optional media and network separately; maintain checkpoint and meaningful retry.
FR-26	UC-03, UC-04	U3, U6	AT-26: choose/skip supported reflection; inspect requests/logs for excluded payloads; text/voice tests await OQ-14.
FR-27	UC-08	U3, U8	AT-27: confirmed erasure request revokes scope immediately; unrelated requester denied; agreed deletion process verified.
FR-28	UC-11	U2, U9	AT-28: Arabic release completeness; nonpublic English fixture shares IDs and progress; explicit fallback.
FR-29	UC-10	U1, U3	AT-29: all five character choices persist through restart/re-pairing and remain editable.
FR-30	UC-10	U1, U2, U4	AT-30: distinguish available, progression-locked, preview, completed, and withdrawn; reject impossible unlocks.
FR-31	UC-02, UC-09	U2, U4	AT-31: all required Dhikr stages exist; door learning completion differs from continuing practice.
FR-32	UC-09	U10	AT-32: create/edit when-then plan and remembering moments; resume without duplicate context records.

Requirement	Use case	Models	Planned acceptance check
FR-33	UC-09	U10	AT-33: pause/resume, missing days, offline duplicate check-in, and timezone edges.
FR-34	UC-09	U10	AT-34: reward once per eligible event; replay and worship counts do not inflate XP.
FR-35	UC-06, UC-11	U2, U7, U9	AT-35: publish locale edition and preview metadata; reject missing Arabic and invalid prerequisite graphs.

Quality coverage

Requirement	Planned check	Baseline dependency
NFR-01	AT-N01: object-level authorization matrix across account, progress, summary, content, and media operations.	Final API surface.
NFR-02	AT-N02: expired/replayed pairing, atomic redemption, rate limit, secure credential storage, token-free logs.	Approved auth mechanism and abuse policy.
NFR-03	AT-N03: network/telemetry/log inspection and deletion/retention exercise.	Retention and release-country decisions.
NFR-04	AT-N04: local restart, server transaction failure, reordered events, duplicate submissions.	Concrete persistence implementation.
NFR-05	AT-N05: measure p95 local interaction timing against proposed 200 ms target.	Agreed representative device.
NFR-06	AT-N06: measure p95 metadata/progress latency against proposed 1 second target.	Agreed concurrency, network, region, and dataset.
NFR-07	AT-N07: screen reader, 200% text, RTL, captions, non-color feedback, reduced motion.	Supported OS/device matrix.
NFR-08	AT-N08: release-build smoke and regression tests on supported platforms and admin browsers.	Launch platform/version decision.
NFR-09	AT-N09: old supported client versus incompatible manifest; safe fallback and preserved progress.	Compatibility policy and template versions.
NFR-10	AT-N10: injected sync/publish faults, correlation IDs, backup restore rehearsal.	Owner, recovery targets, backup/retention policy.
NFR-11	AT-N11: source/rights and age-suitability review exists for every published revision.	Appointed editorial owner and completed content.
NFR-12	AT-N12: observed child usability sessions across the age range.	Sample/recruitment plan and agreed threshold.
NFR-13	AT-N13: Arabic/RTL and English/LTR fixtures, text growth, media alternatives, locale fallback, unchanged progress.	Locale contract and release-language policy.

Requirement	Planned check	Baseline dependency
NFR-14	AT-N14: duplicate check-ins/rewards, timezones, content revisions, and paused practice.	Confirmed check-in/reward and calendar rules.

Confirmed product needs mapped to the specification

Product decision	Coverage
C-01: ages 10-14	FR-02, NFR-11, NFR-12
C-02: mother's own journey and support	FR-05, FR-14-FR-16
C-03: child independence / optional mother	FR-10, FR-11, FR-16, BR-01
C-04: Dhikr replaces former first journey	SRS scope, FR-06, UC-02
C-05: light fantasy / everyday dialogue	FR-06, UC-02, editorial review
C-06: Expo React Native	SRS interfaces, NFR-08, architecture / U8
C-07: separate devices	FR-03-FR-05, FR-12, U5, U6
C-08: admin content updates	FR-17-FR-24, UC-06/UC-07
C-09: complete single-door MVP	SRS scope, FR-06, AT-17 tests later extensibility
C-10: presentation only	Executive brief and source limitations; production content remains a dependency
C-11: documents before UI	This pack; no UI implementation or visual design is performed
C-12: Arabic first, English-ready structure	FR-28, FR-35, NFR-13, UC-11, U9
C-13: child character creation	FR-29, UC-10, U3
C-14: locked-door map	FR-06, FR-30, UC-10, U2/U4

Documentation verification versus product verification

Documentation checks validate file links, unique IDs, coverage, and renderable diagram syntax. They cannot prove that an implementation is secure, a child understands the experience, the religious content is correct, or the service meets latency targets. Those checks occur after the relevant content and software exist.

09 Review and document control

Review sequence

1. Confirm the first-release scope and the proposed family/account model.
2. Agree mother visibility, child privacy, and progress expectations across devices.
3. Approve supported content templates and assign editorial ownership.
4. Resolve launch-platform, connectivity, operating, and data-retention decisions.
5. Baseline the SRS, then update the use cases, UML, and acceptance mapping together.

Change control

Keep requirement IDs stable. Record an explicit decision before promoting a proposal to the approved baseline. Revise related diagrams and acceptance checks when behavior changes. A routine content edit must not silently rewrite historical learner completion.

Source material

- Presentation 13.pptx: supplied conceptual project presentation.
- Expanded project notes: supplied written material about the learning themes and journey.
- Client revision, 29 September 2026: ages 10-14, complete Dhikr door, character creation, locked-door map, and Arabic-first localization. Written/voice reflection is a pending scope choice.
- Product-planning decisions: confirmed audience, mother/child participation model, first topic, visual direction, framework, separate devices, and admin content updates.

Verification status

The source pack contains 35 functional requirements, 14 nonfunctional requirements, 15 business rules, twelve detailed use cases, and ten UML diagrams. Requirement identifiers and traceability coverage were checked, and the diagrams were rendered from their editable sources. These documentation checks do not substitute for application, security, accessibility, content, or usability acceptance testing.

Review record

Record agreed decisions and revisions here before establishing an approved baseline.

Field	Review entry
Document version	0.2 - client review draft
Review date	To be recorded
Reviewer / role	To be recorded
Decisions accepted	Reference C-, P-, OQ-, or requirement IDs
Changes requested	Reference requirement IDs and affected models
Baseline status	Not yet approved